

Презентация по лабораторной работе 5

Конфигурирование VLAN

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1.1 Докладчик

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- Российский университет дружбы народов им. П. Лумумбы

2 Вводная часть

2.1 Цели и задачи

- Получить основные навыки по настройке VLAN на коммутаторах сети.

3 Выполнение заданий

3.1 слайд 1

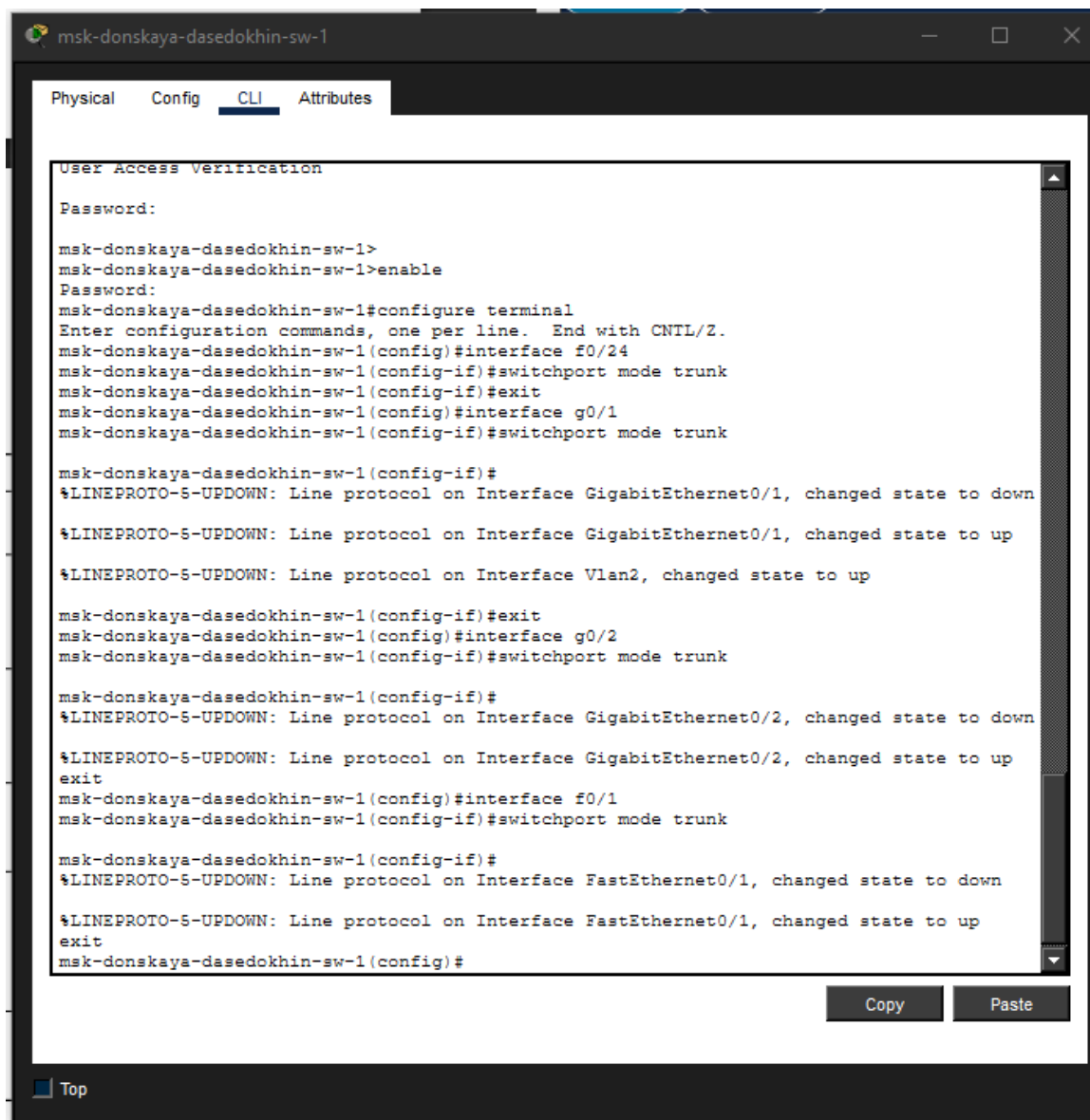
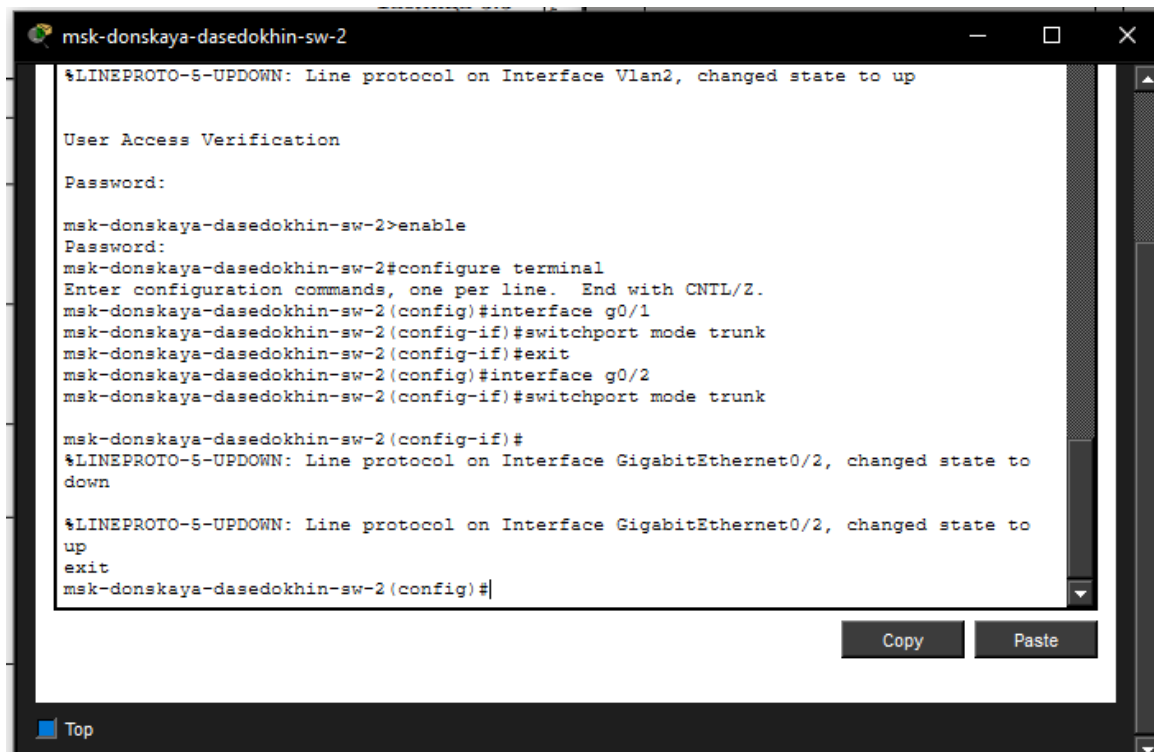


Рисунок 3.1: Настройка Trunk-порта коммутатора `msk-donskaya-dasedokhin-sw-1`

3.2 слайд 2



```
msk-donskaya-dasedokhin-sw-2
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

User Access Verification

Password:

msk-donskaya-dasedokhin-sw-2>enable
Password:
msk-donskaya-dasedokhin-sw-2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-dasedokhin-sw-2(config)#interface g0/1
msk-donskaya-dasedokhin-sw-2(config-if)#switchport mode trunk
msk-donskaya-dasedokhin-sw-2(config-if)#exit
msk-donskaya-dasedokhin-sw-2(config)#interface g0/2
msk-donskaya-dasedokhin-sw-2(config-if)#switchport mode trunk

msk-donskaya-dasedokhin-sw-2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to
down

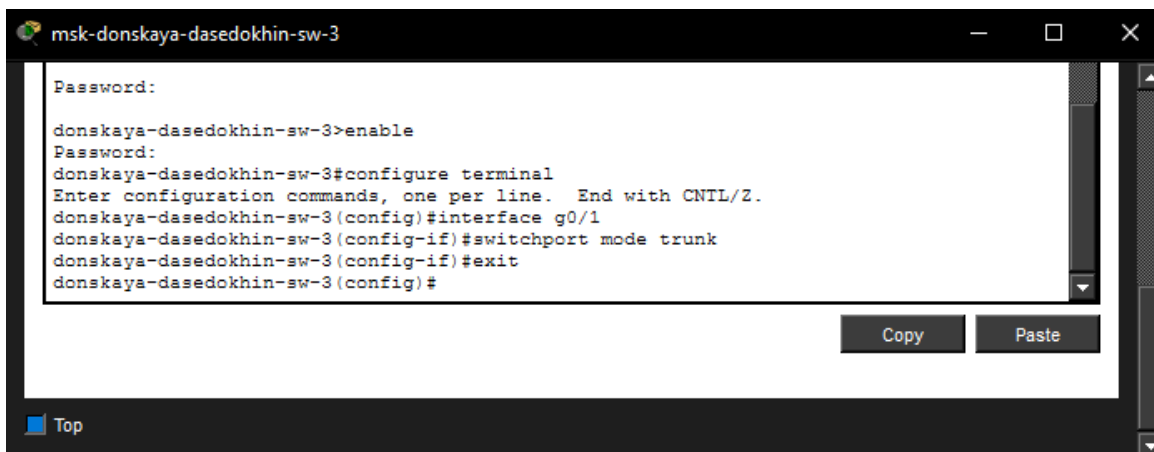
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to
up
exit
msk-donskaya-dasedokhin-sw-2(config)#|
```

Copy Paste

Top

Рисунок 3.2: Настройка Trunk-порта коммутатора msk-donskaya-dasedokhin-sw-2

3.3 слайд 3



```
msk-donskaya-dasedokhin-sw-3
Password:

donskaya-dasedokhin-sw-3>enable
Password:
donskaya-dasedokhin-sw-3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
donskaya-dasedokhin-sw-3(config)#interface g0/1
donskaya-dasedokhin-sw-3(config-if)#switchport mode trunk
donskaya-dasedokhin-sw-3(config-if)#exit
donskaya-dasedokhin-sw-3(config)#
```

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Top

Рисунок 3.3: Настройка Trunk-порта коммутатора msk-donskaya-dasedokhin-sw-3

3.4 слайд 4

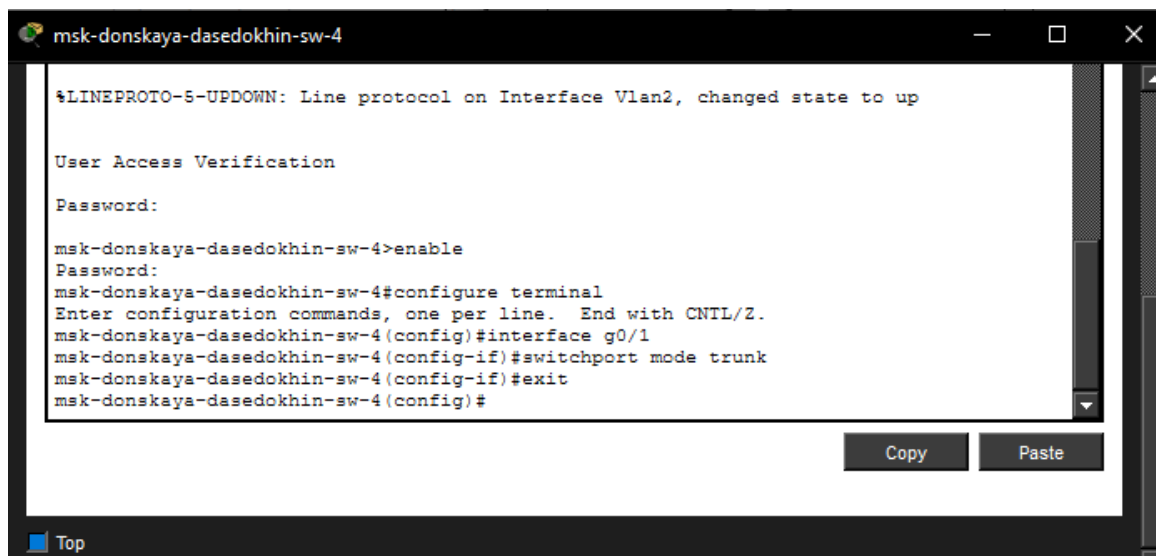


Рисунок 3.4: Настройка Trunk-порта коммутатора msk-donskaya-dasedokhin-sw-4

3.5 слайд 5

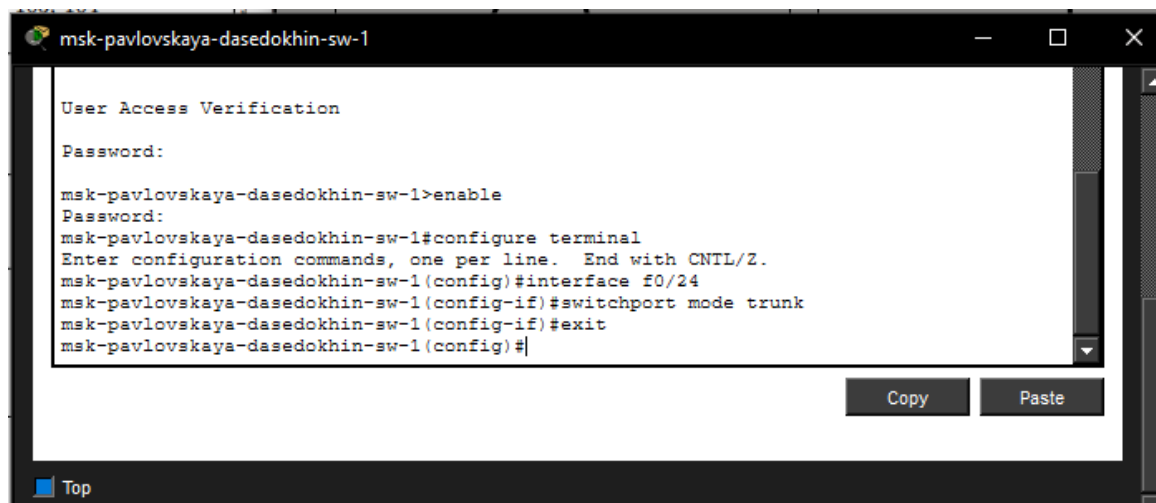
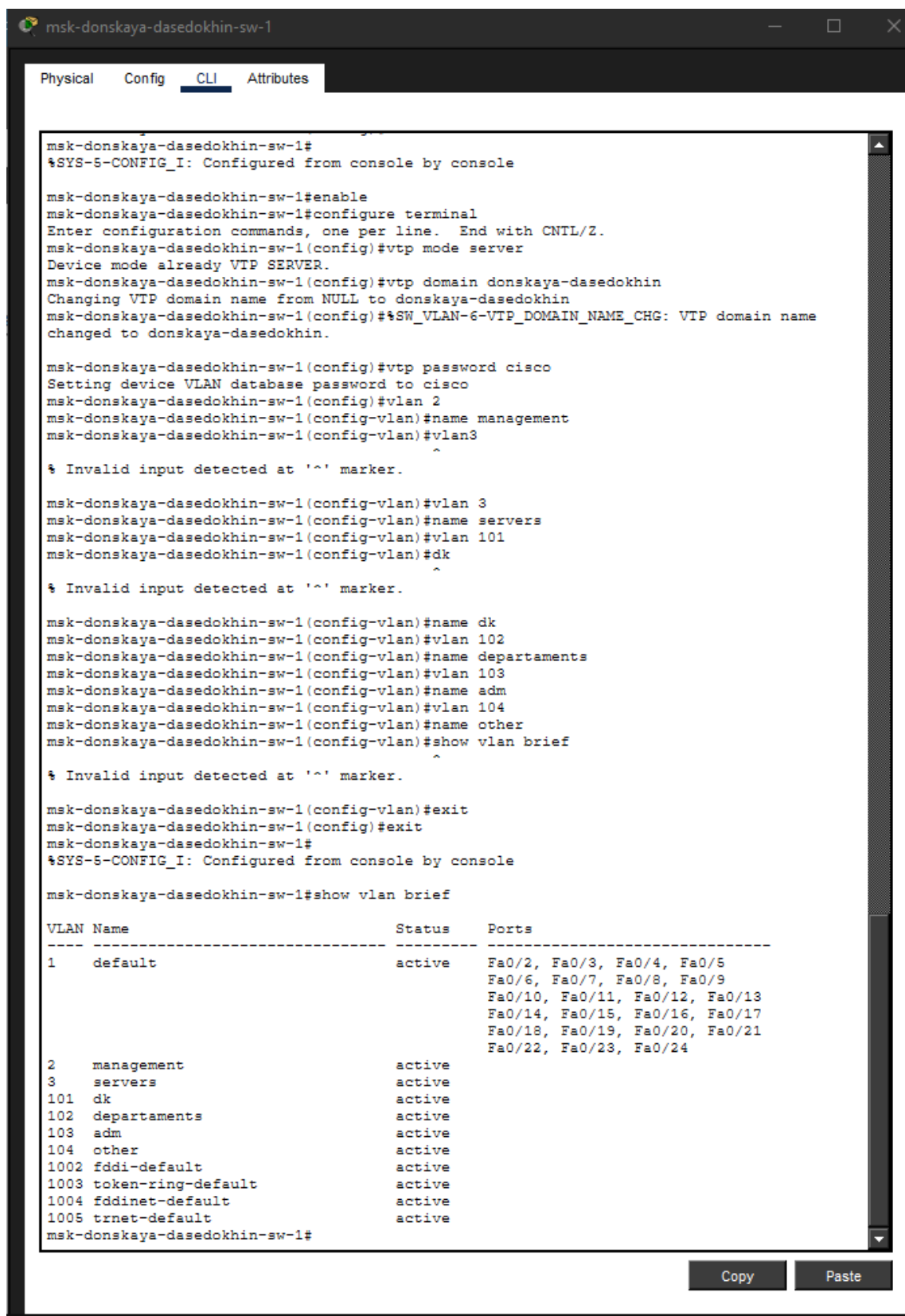


Рисунок 3.5: Настройка Trunk-порта коммутатора msk-pavlovskaya-dasedokhin-sw-1

3.6 слайд 6



```
msk-donskaya-dasedokhin-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-dasedokhin-sw-1#enable
msk-donskaya-dasedokhin-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-dasedokhin-sw-1(config)#vtp mode server
Device mode already VTP SERVER.
msk-donskaya-dasedokhin-sw-1(config)#vtp domain donsкаya-dasedokhin
Changing VTP domain name from NULL to donsкаya-dasedokhin
msk-donskaya-dasedokhin-sw-1(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name
changed to donsкаya-dasedokhin.

msk-donskaya-dasedokhin-sw-1(config)#vtp password cisco
Setting device VLAN database password to cisco
msk-donskaya-dasedokhin-sw-1(config)#vlan 2
msk-donskaya-dasedokhin-sw-1(config-vlan)#name management
msk-donskaya-dasedokhin-sw-1(config-vlan)#vlan3
^
% Invalid input detected at '^' marker.

msk-donskaya-dasedokhin-sw-1(config-vlan)#vlan 3
msk-donskaya-dasedokhin-sw-1(config-vlan)#name servers
msk-donskaya-dasedokhin-sw-1(config-vlan)#vlan 101
msk-donskaya-dasedokhin-sw-1(config-vlan)#dk
^
% Invalid input detected at '^' marker.

msk-donskaya-dasedokhin-sw-1(config-vlan)#name dk
msk-donskaya-dasedokhin-sw-1(config-vlan)#vlan 102
msk-donskaya-dasedokhin-sw-1(config-vlan)#name departaments
msk-donskaya-dasedokhin-sw-1(config-vlan)#vlan 103
msk-donskaya-dasedokhin-sw-1(config-vlan)#name adm
msk-donskaya-dasedokhin-sw-1(config-vlan)#vlan 104
msk-donskaya-dasedokhin-sw-1(config-vlan)#name other
msk-donskaya-dasedokhin-sw-1(config-vlan)#show vlan brief
^
% Invalid input detected at '^' marker.

msk-donskaya-dasedokhin-sw-1(config-vlan)#exit
msk-donskaya-dasedokhin-sw-1(config)#exit
msk-donskaya-dasedokhin-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-dasedokhin-sw-1#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                           Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                           Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                           Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                           Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                           Fa0/22, Fa0/23, Fa0/24
2    management             active
3    servers                 active
101  dk                       active
102  departaments             active
103  adm                      active
104  other                    active
1002 fddi-default            active
1003 token-ring-default      active
1004 fddinet-default          active
1005 trnet-default            active
msk-donskaya-dasedokhin-sw-1#
```

Рисунок 3.6: Прописываем msk-donskaya-dasedokhin-sw-1 как VTP-сервер и номера, названия VLAN

3.7 слайд 7

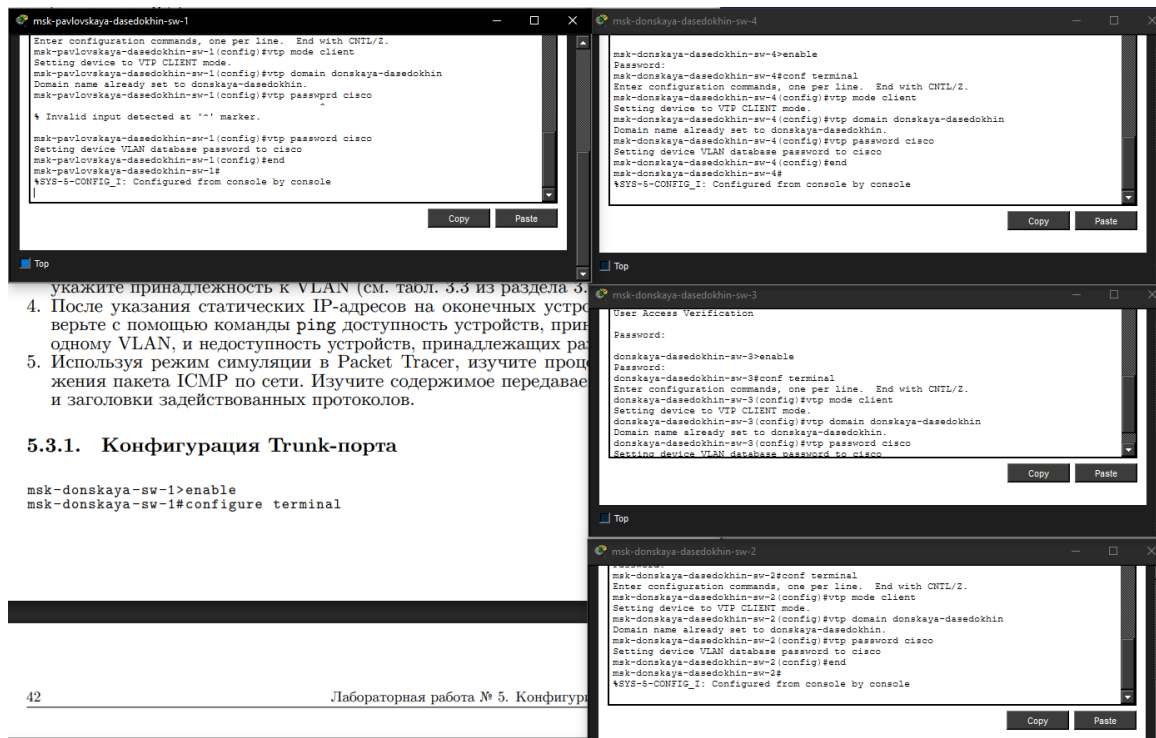
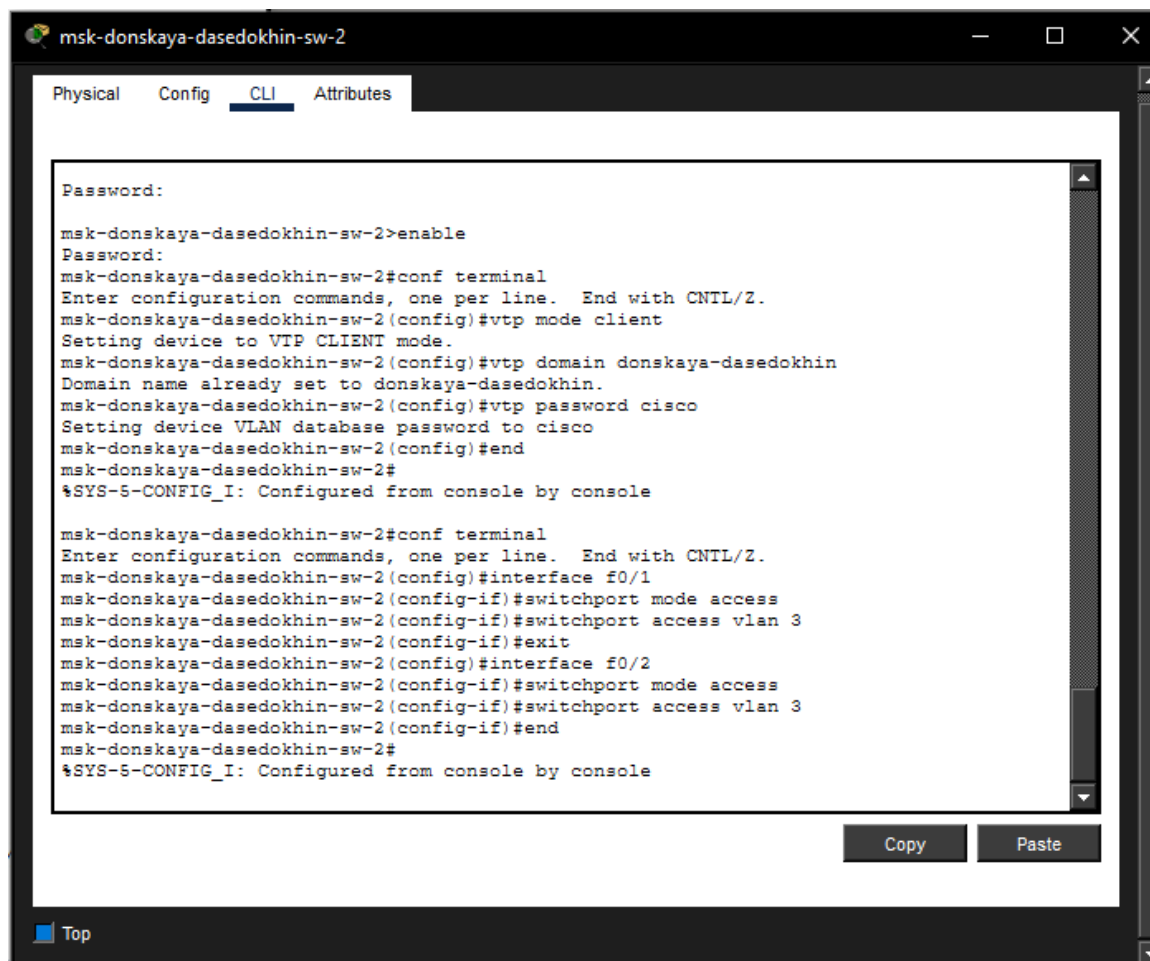


Рисунок 3.7: Наситрока msk-donskaya-sw-2 — mskdonskaya-sw-4, msk-pavlovskaya-sw-1 как VTP-клиенты

3.8 слайд 8



The screenshot shows a network configuration window titled "msk-donskaya-dasedokhin-sw-2". It has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The CLI window displays the following commands and output:

```
msk-donskaya-dasedokhin-sw-2>enable
msk-donskaya-dasedokhin-sw-2#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-dasedokhin-sw-2(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-dasedokhin-sw-2(config)#vtp domain donskeya-dasedokhin
Domain name already set to donskeya-dasedokhin.
msk-donskaya-dasedokhin-sw-2(config)#vtp password cisco
Setting device VLAN database password to cisco
msk-donskaya-dasedokhin-sw-2(config)#end
msk-donskaya-dasedokhin-sw-2#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-dasedokhin-sw-2#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-dasedokhin-sw-2(config)#interface f0/1
msk-donskaya-dasedokhin-sw-2(config-if)#switchport mode access
msk-donskaya-dasedokhin-sw-2(config-if)#switchport access vlan 3
msk-donskaya-dasedokhin-sw-2(config-if)#exit
msk-donskaya-dasedokhin-sw-2(config)#interface f0/2
msk-donskaya-dasedokhin-sw-2(config-if)#switchport mode access
msk-donskaya-dasedokhin-sw-2(config-if)#switchport access vlan 3
msk-donskaya-dasedokhin-sw-2(config-if)#end
msk-donskaya-dasedokhin-sw-2#
%SYS-5-CONFIG_I: Configured from console by console
```

At the bottom of the CLI window, there are "Copy" and "Paste" buttons. Below the CLI window, there is a "Top" button.

Рисунок 3.8: Принадлежность к VLAN для msk-donskaya-dasedokhin-sw-2

3.9 слайд 9

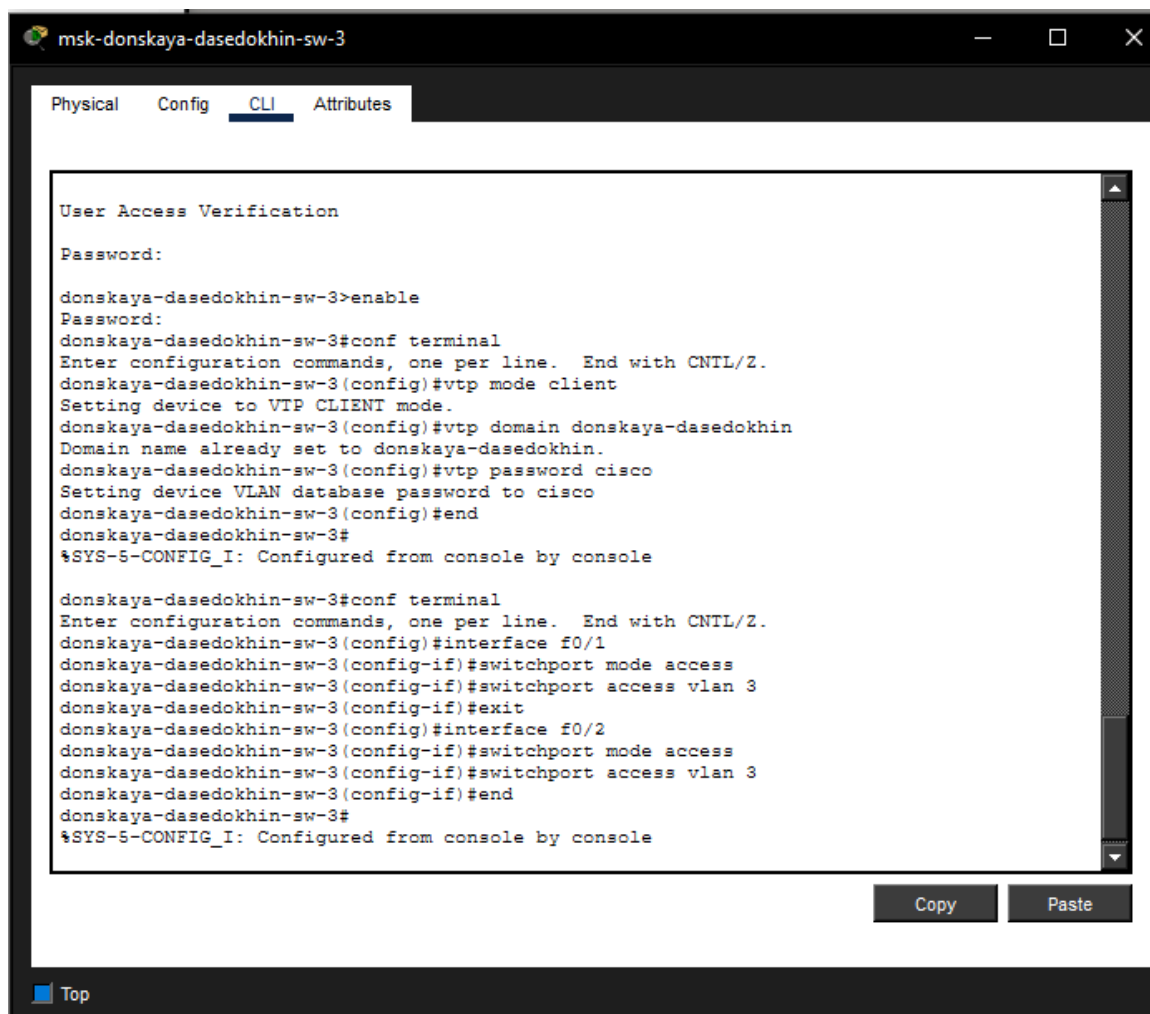
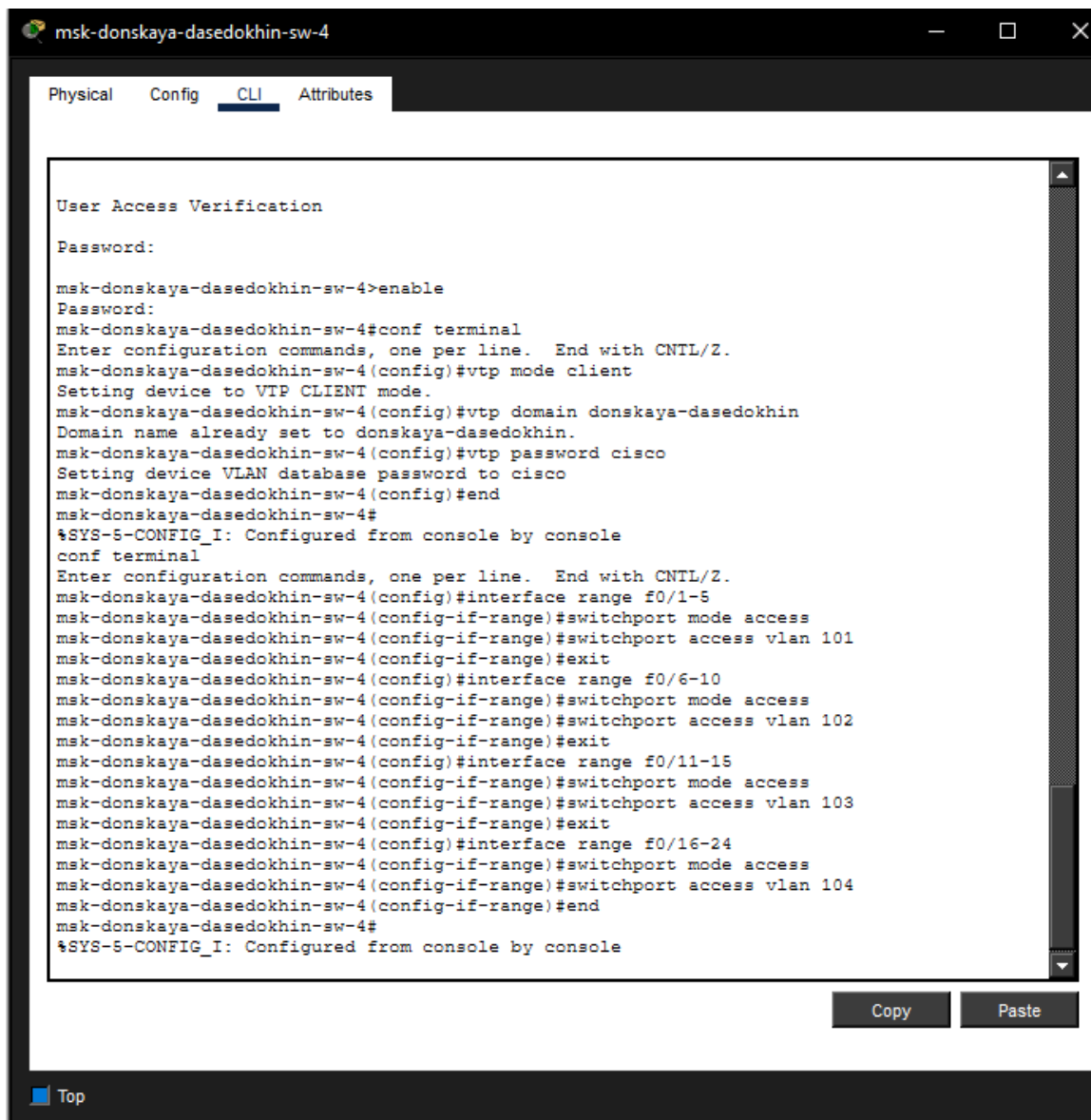


Рисунок 3.9: Принадлежность к VLAN для msk-donskaya-dasedokhin-sw-3

3.10 слайд 10



The screenshot shows a network configuration window titled "msk-donskaya-dasedokhin-sw-4". It has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The main area displays a terminal session for "User Access Verification". The session starts with a password prompt, followed by the command "enable". Then, "conf terminal" is entered to start configuration mode. The configuration includes setting the VTP mode to client, domain to "donskaya-dasedokhin", and password to "cisco". After exiting configuration mode, the user enters "conf terminal" again to configure interfaces. The configuration consists of three interface ranges: f0/1-5, f0/6-10, and f0/11-15, each configured as an access port and assigned to VLANs 101, 102, and 103 respectively. The final interface range is f0/16-24, configured as an access port and assigned to VLAN 104. The session ends with a confirmation message from the system.

```
User Access Verification

Password:

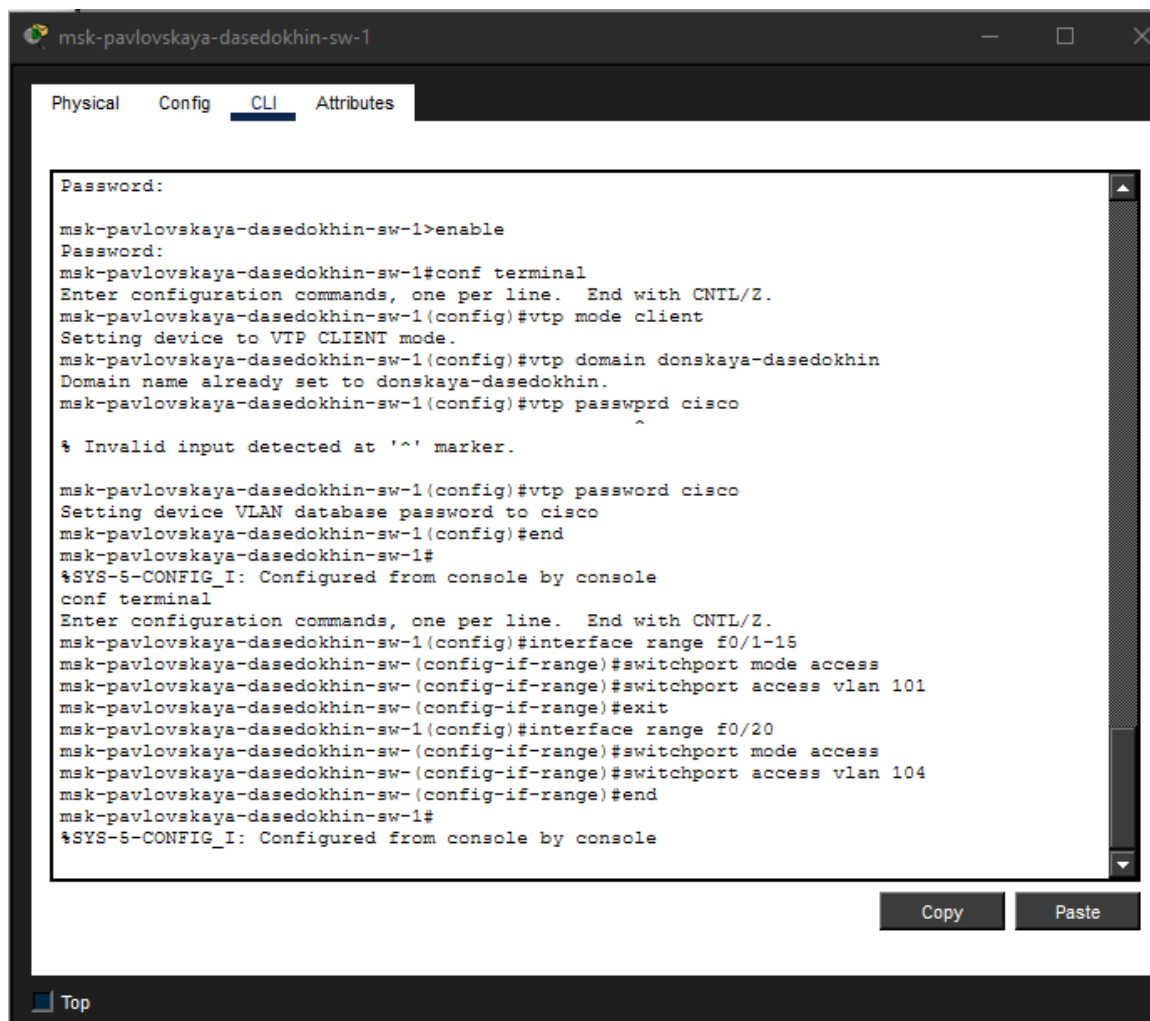
msk-donskaya-dasedokhin-sw-4>enable
Password:
msk-donskaya-dasedokhin-sw-4#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-dasedokhin-sw-4(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-dasedokhin-sw-4(config)#vtp domain donsкаya-dasedokhin
Domain name already set to donsкаya-dasedokhin.
msk-donskaya-dasedokhin-sw-4(config)#vtp password cisco
Setting device VLAN database password to cisco
msk-donskaya-dasedokhin-sw-4(config)#end
msk-donskaya-dasedokhin-sw-4#
%SYS-5-CONFIG_I: Configured from console by console
conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-dasedokhin-sw-4(config)#interface range f0/1-5
msk-donskaya-dasedokhin-sw-4(config-if-range)#switchport mode access
msk-donskaya-dasedokhin-sw-4(config-if-range)#switchport access vlan 101
msk-donskaya-dasedokhin-sw-4(config-if-range)#exit
msk-donskaya-dasedokhin-sw-4(config)#interface range f0/6-10
msk-donskaya-dasedokhin-sw-4(config-if-range)#switchport mode access
msk-donskaya-dasedokhin-sw-4(config-if-range)#switchport access vlan 102
msk-donskaya-dasedokhin-sw-4(config-if-range)#exit
msk-donskaya-dasedokhin-sw-4(config)#interface range f0/11-15
msk-donskaya-dasedokhin-sw-4(config-if-range)#switchport mode access
msk-donskaya-dasedokhin-sw-4(config-if-range)#switchport access vlan 103
msk-donskaya-dasedokhin-sw-4(config-if-range)#exit
msk-donskaya-dasedokhin-sw-4(config)#interface range f0/16-24
msk-donskaya-dasedokhin-sw-4(config-if-range)#switchport mode access
msk-donskaya-dasedokhin-sw-4(config-if-range)#switchport access vlan 104
msk-donskaya-dasedokhin-sw-4(config-if-range)#end
msk-donskaya-dasedokhin-sw-4#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

Top

Рисунок 3.10: Принадлежность к VLAN для msk-donskaya-dasedokhin-sw-4

3.11 слайд 11



```
msk-pavlovskaya-dasedokhin-sw-1
Physical Config CLI Attributes

Password:

msk-pavlovskaya-dasedokhin-sw-1>enable
Password:
msk-pavlovskaya-dasedokhin-sw-1#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-pavlovskaya-dasedokhin-sw-1(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-pavlovskaya-dasedokhin-sw-1(config)#vtp domain donskeya-dasedokhin
Domain name already set to donskeya-dasedokhin.
msk-pavlovskaya-dasedokhin-sw-1(config)#vtp passwprd cisco
^
% Invalid input detected at '^' marker.

msk-pavlovskaya-dasedokhin-sw-1(config)#vtp password cisco
Setting device VLAN database password to cisco
msk-pavlovskaya-dasedokhin-sw-1(config)#end
msk-pavlovskaya-dasedokhin-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-pavlovskaya-dasedokhin-sw-1(config)#interface range f0/1-15
msk-pavlovskaya-dasedokhin-sw-1(config-if-range)#switchport mode access
msk-pavlovskaya-dasedokhin-sw-1(config-if-range)#switchport access vlan 101
msk-pavlovskaya-dasedokhin-sw-1(config-if-range)#exit
msk-pavlovskaya-dasedokhin-sw-1(config)#interface range f0/20
msk-pavlovskaya-dasedokhin-sw-1(config-if-range)#switchport mode access
msk-pavlovskaya-dasedokhin-sw-1(config-if-range)#switchport access vlan 104
msk-pavlovskaya-dasedokhin-sw-1(config-if-range)#end
msk-pavlovskaya-dasedokhin-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Рисунок 3.11: Принадлежность к VLAN для msk-pavlovskaya-dasedokhin-sw-1

3.12 слайд 12

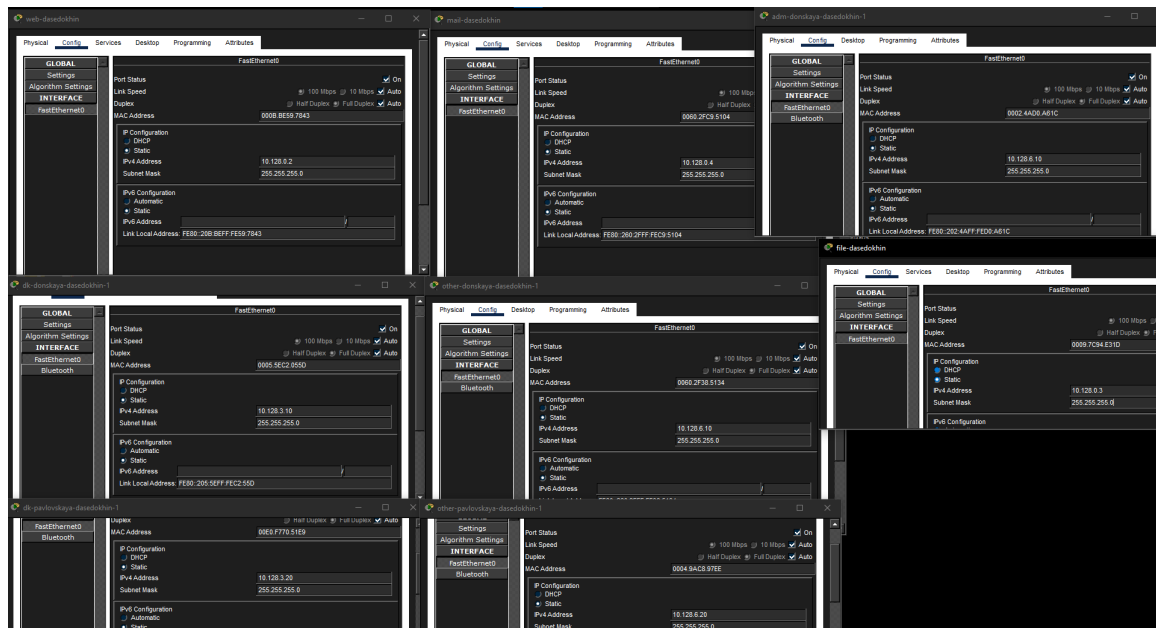
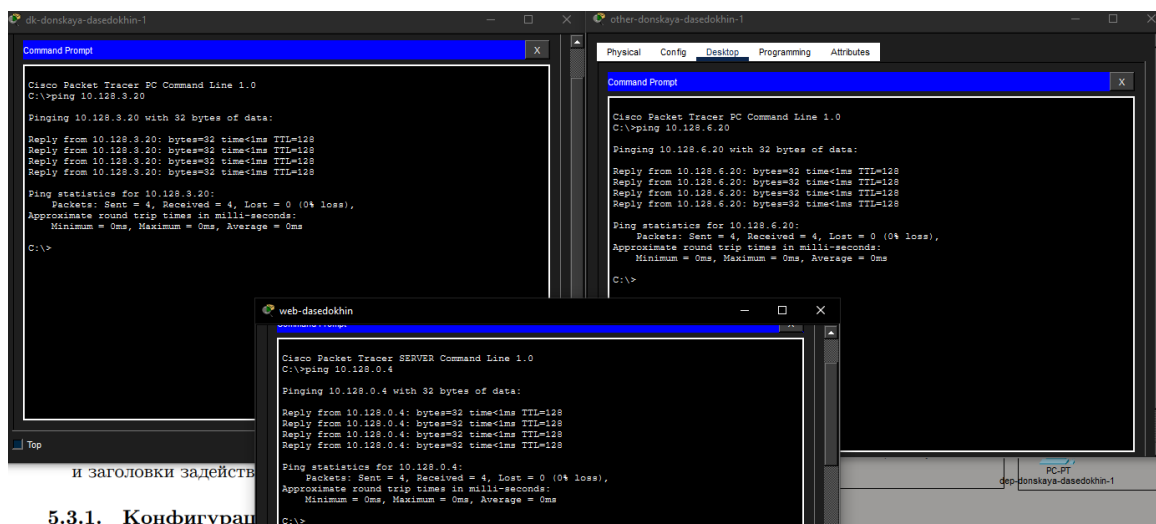


Рисунок 3.12: Статические IP-адреса для оконечных устройств

3.13 слайд 13



и заголовки задействованы

5.3.1. Конфигурация

Рисунок 3.13: Пинг устройств принадлежащих одному VLAN

3.14 слайд 14

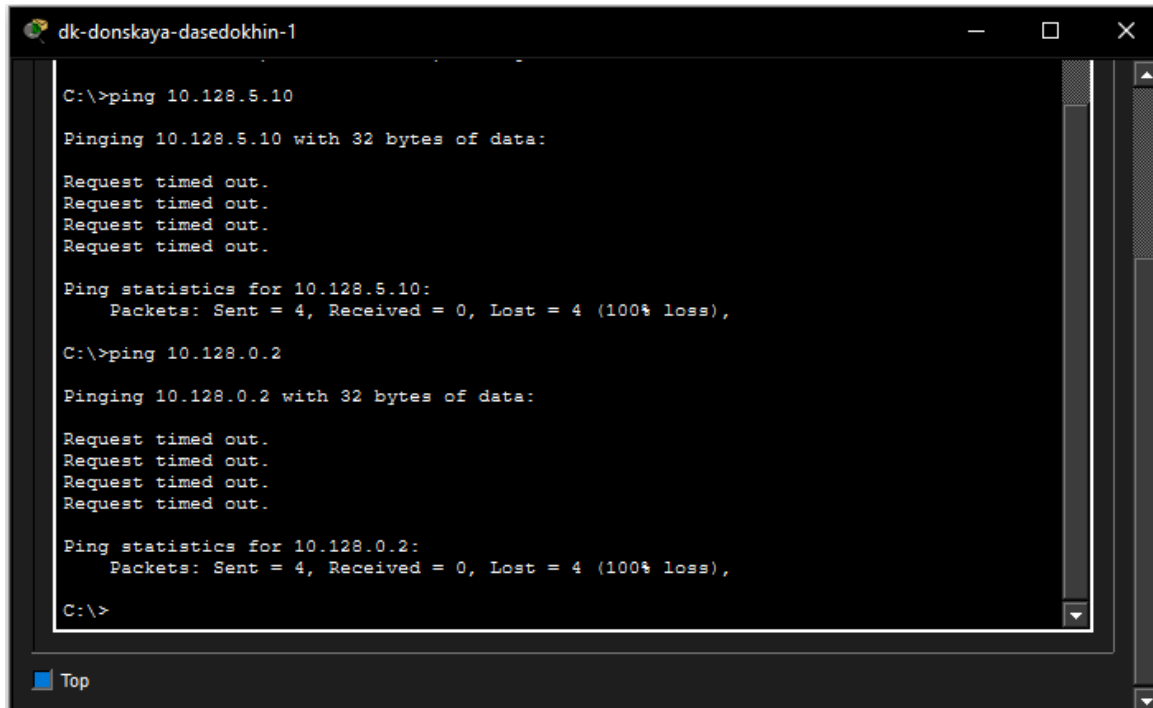


Рисунок 3.14: Пинг устройств принадлежащих разным VLAN

3.15 слайд 15

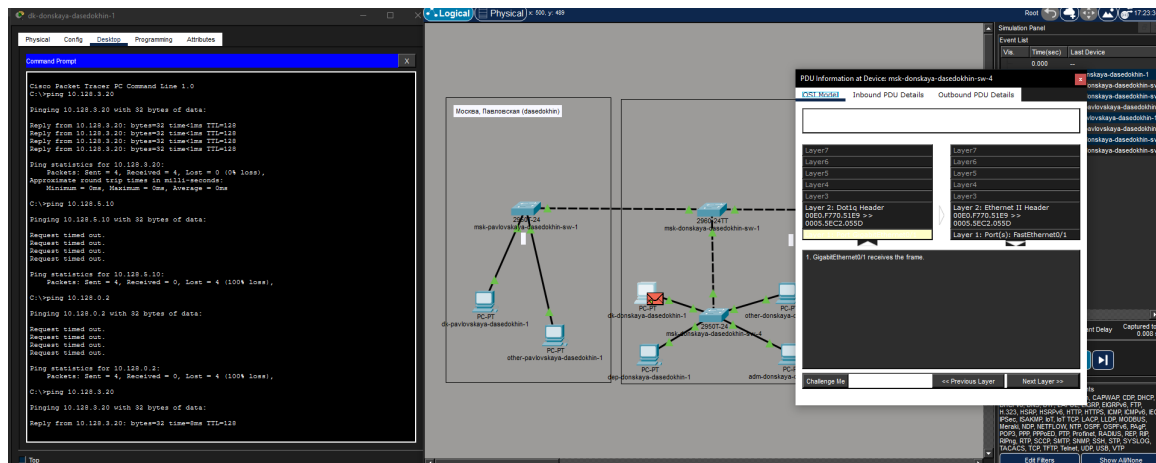


Рисунок 3.15: Просмотр пути пакета ICMP, изучение информации о содержимом пакета

4 Выводы

4.1 слайд 1

Я Получил основные навыки по настройке VLAN на коммутаторах сети.